CCCA Meta-Atlas — Cell-State Manifold Analysis

Griffiths et al. 2021 · Breast Cancer · 10x + iCell8 · 108,849 malignant cells on the trajectory

GO / NO-GO GATES

Gate 1 — integration & continuity — GO

Samples/patients intermix per-neighbourhood on ONE connected manifold. Harmony(sample) same-sample neighbour fraction 0.91 -> 0.17 (theta=5); iLISI 1.0 -> 13.4 of 67 samples. ~1,100 detached cells (1.0%) removed as islands; largest component retained. Mixed AND continuous.

Gate 2 — embryonic module geography — GO (non-branching)

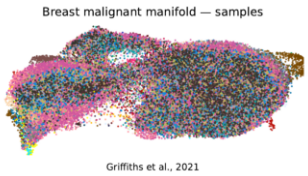
Single Gast/pSom -> Crest axis, no hub diverging into two arms. Organized Gast<->pSom cross-repression (smoothed r = -0.42); Crest an inverse terminal gradient (Crest vs pSom r = -0.76; NC-high decile at pseudotime 0.77; joint pSom&Crest enrichment 0.31 < 1). Curve fits clean (2 tips, 0 forks, fold-free); PPT tree only over-fragments (5-13 forks). Crest does NOT enrich a Gast arm (no arms) — no flag.

Gate 3 — hub cross-repression — NA

No bifurcation (non-branching geometry) -> branching-only Gate 3 not evaluated; 04_correlation / 05_modules not produced (spec 6).

STEP FIGURES

01 · samples (Gate 1)



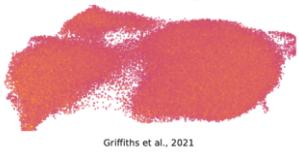
02 · pseudotime



03 · Gastrulation



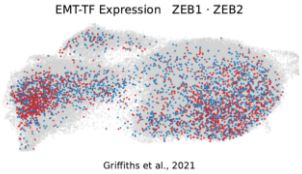
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TF overlay



PARAMETER CHOICES & RATIONALE

Min-cell threshold >= 100 (derived): per-sample counts smooth above the floor — no dominant spillover cliff (only gap is the 5->1 doublet tail) — so the depth-invariant floor applies. 67/95 samples, 110,195 cells.

Two technology leaves (10X 110,568 + iCell8 804 malignant) loaded per spec 4 and concatenated on 15,691 shared genes.

Embedding n_pcs=30, Harmony(sample) theta=5 (strongest integration before the mixing plateau), spread=1.3, k=30.

Trajectory scFates curve (Nodes=20) — matched to non-branching topology; fold self-approach 2.95 (fold-free).

Compartment 246-cell lobe (100% iCell8, proliferation 0.70) scoped out as a technology/cycling confound.

Root curve tip 0 (Gast/pSom-high, Crest-low = developmental origin); axis cell-cycle-free (pt vs cycling r=-0.04).

Geometry Linear-pSom/Crest (global means Gast -0.021, pSom -0.052, Crest +0.021).

Reproducibility Harmony KMeans is not seed-deterministic: after-island count jitters ~0.1% (109,095 +/- ~100); threshold, theta, gates and all downstream decisions are invariant to it.

EMT-TFs panel Snail/Zeb/Twist/FOXC/PRRX gated on detectable expression; ZEB1, PRRX1, ZEB2 expressed but essentially flat along pseudotime (all |rho|<0.05) — no early/late axis; overlay the two most-expressed (ZEB1, ZEB2) non-directionally (VIM excluded — mesenchymal marker, not a TF).

VERDICT

The embryonic-program pattern reproduces as the non-branching case: one continuous, sample-integrated gastrulation/somitogenesis -> neural-crest axis with organized Gast<->pSom cross-repression and terminal Crest — no Gast/pSom hub bifurcating into two competing arms. Gates: G1 GO · G2 GO (non-branching) · G3 NA.